

Autonomous Space Debris Collision Avoidance: A Data-Driven Reinforcement Learning Approach in the OpenEnv Framework

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Abstract—The exponential increase in orbital space debris presents a critical threat to active satellites and human spaceflight. Managing conjunction events traditionally relies on human-in-the-loop decision-making, which struggles to scale with the sheer volume of tracked objects in Low Earth Orbit (LEO). In this paper, we present a data-driven, autonomous approach to spacecraft collision avoidance utilizing Large Language Models (LLMs) embedded within a Reinforcement Learning (RL) loop, simulated via the Orbits OpenEnv framework. By analyzing historical conjunction data, we derive operational priors that inform an iterative self-improvement strategy for the autonomous agent. Our methodology focuses on optimizing maneuvering decisions under high-uncertainty conditions while preserving mission integrity and adhering to stringent tracking budgets. The experimental results, generated over multiple inference rounds, demonstrate the agent’s evolving capacity to balance fuel consumption with risk reduction across varying task difficulties. Through comprehensive exploratory data analysis and simulated RL episodes, this report highlights the feasibility of LLM-driven agents in complex, multi-objective aerospace environments.

Index Terms—Space Debris, Collision Avoidance, Reinforcement Learning, Large Language Models, OpenEnv, Data Analytics, Orbital Mechanics.

1 Introduction

Space operations in the 21st century are increasingly threatened by the proliferation of orbital debris. Since the launch of Sputnik in 1957, thousands of defunct satellites, spent rocket bodies, and fragmentation pieces have accumulated in Earth’s orbit. Traveling at orbital velocities exceeding 7 km/s, even millimeter-sized particles carry kinetic energy sufficient to critically damage operational spacecraft. The Kessler Syndrome—a theoretical scenario in which the density of objects in Low Earth Orbit (LEO) becomes high enough that collisions between objects cause a cascade—poses an existential risk to satellite constellations, global communication networks, and international security.

Traditionally, space agencies and commercial operators rely on ground-based radar and optical tracking systems to catalog orbital objects and predict close approaches, known as

conjunctions. When a conjunction event exceeds a predefined probability of collision (P_c) threshold, human operators are alerted to design and execute a collision avoidance maneuver (CAM). However, as mega-constellations are deployed and sensor resolution improves, the frequency of conjunction alerts has skyrocketed, overwhelming human operators.

This necessitates the development of autonomous systems capable of real-time conjunction assessment and maneuver planning. Reinforcement Learning (RL) has emerged as a promising paradigm for solving complex sequential decision-making problems. Yet, classical RL models often struggle with the sparse rewards, high-dimensional state spaces, and partial observability inherent to orbital mechanics. Furthermore, training bespoke RL agents requires millions of simulated episodes, making them computationally expensive.

In recent years, Large Language Models (LLMs) have demonstrated zero-shot and few-shot reasoning capabilities across diverse domains. By prompting an LLM with structured state representations and operational priors, it can act as a generalized heuristic engine. This paper investigates the efficacy of LLM-based autonomous agents for space debris collision avoidance within the Orbits OpenEnv simulation framework.

The primary contributions of this paper are:

- 1) A comprehensive Exploratory Data Analysis (EDA) of orbital debris datasets, extracting critical priors regarding object distributions and uncertainty profiles.
- 2) The implementation of an autonomous LLM-driven agent that navigates the Orbits OpenEnv framework, executing maneuvers and requesting tracking updates under constraints.
- 3) An iterative self-improvement loop that allows the agent to refine its strategies based on historical performance metrics.
- 4) Extensive empirical evaluation detailing the agent’s score progression, reward generation, and action selection distributions across multiple levels of difficulty.

The remainder of this paper is structured as follows. Section II provides the background on orbital mechanics and the space debris environment. Section III details the OpenEnv framework architecture. Section IV presents our data analytics and derived priors. Section V outlines the autonomous agent methodology. Section VI describes the experimental setup. Section VII presents the results and analysis, followed by discussions in Section VIII, future work in Section IX, and conclusions in Section X.

2 Background and Motivation

2.1 Orbital Mechanics Fundamentals

Spacecraft in Earth’s orbit follow Keplerian trajectories, which are defined by six classical orbital elements: semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of perigee, and true anomaly. While these elements describe the macroscopic orbit, collision avoidance is primarily concerned with the relative motion between the primary spacecraft and a secondary object (the chaser or debris) during a brief window around the Time of Closest Approach (TCA).

The relative geometry is typically modeled in a localized Cartesian coordinate system attached to the primary spacecraft, known as the Radial-Transverse-Normal (RTN) or Radial-Along-Track-Cross-Track (RIC) frame.

- Radial (R): Directed from the center of the Earth through the spacecraft.
- Along-Track (T or I): Directed perpendicular to the radial vector, generally in the direction of motion.
- Normal (N or C): Directed perpendicular to the orbital plane, completing the right-handed triad.

Maneuvers executed along these axes have distinct effects. Radial thrusts primarily change the orbital period and eccentricity. Along-track thrusts are the most efficient for changing the semi-major axis (and thus the altitude at the opposite side of the orbit). Normal thrusts change the orbital inclination. Understanding these effects is vital for an autonomous agent to select the most fuel-efficient maneuver that maximizes the miss distance.

2.2 The Space Debris Environment

The space debris population is highly heterogeneous. It includes:

- Intact Objects: Dead satellites and abandoned rocket bodies.
- Fragmentation Debris: Shards created by accidental explosions, anti-satellite weapon (ASAT) tests, or historical collisions (e.g., the 2009 Iridium-Cosmos collision).
- Mission-Related Debris: Objects released intentionally during operations, such as lens covers or deployment mechanisms.

Because radar cross-sections vary wildly, tracking these objects is fraught with uncertainty. The Combined Space Operations Center (CSpOC) maintains the public satellite catalog (SATCAT), but tracking precision degrades rapidly due to atmospheric drag fluctuations, solar radiation pressure, and the inherent limits of ground-based sensors.

2.3 Collision Avoidance Under Uncertainty

The fundamental challenge of collision avoidance is decision-making under uncertainty. A high probability of collision (P_c) might simply be an artifact of high covariance (uncertainty) in the tracking data. Conversely, a low P_c might obscure a genuine threat if the object’s position is poorly constrained.

Operators must balance three competing objectives: 1. Safety: Keeping the P_c below an acceptable threshold (typically 10^{-4}). 2. Resource Conservation: Propellant (Delta-V) is strictly limited. Exhausting fuel reduces the operational lifespan of the satellite. 3. Mission Integrity: Maneuvers perturb the satellite from its nominal orbit. Drifting too far compromises mission objectives (e.g., Earth observation targeting or communication links).

3 The OpenEnv Framework

To systematically evaluate autonomous agents, we utilize the Orbits OpenEnv framework. This environment abstracts the continuous, computationally heavy physics of orbital propagation into a discrete, structured Markov Decision Process (MDP).

3.1 Architecture and Abstraction

The OpenEnv framework utilizes a client-server architecture. The core simulation engine, written in Python, calculates deterministic outcomes for stochastic inputs, ensuring reproducible evaluations. It is wrapped in a FastAPI web server, enabling agents (the clients) to interact via standard REST APIs. This decouples the agent’s logic from the physical simulation, standardizing the evaluation pipeline.

Instead of propagating high-fidelity Cartesian state vectors, the simulator models conjunction events abstractly. Each event maintains a baseline collision probability, a miss distance, and an uncertainty metric. Maneuvers act geometrically; the simulator maps specific action types (e.g., Radial Maneuver) to reduction coefficients depending on the specific geometry of the approaching debris.

3.2 State Representation and Action Space

The environment provides the agent with an Observation at each step t . The observation space O_t includes:

- Step Index (t) and Horizon: The remaining turns before the episode ends.
- Resource Metrics: Remaining fuel and tracking budget.
- Mission Offsets: The cumulative drift across the Radial, Along-Track, and Normal axes.
- Visible Events: A list of the most threatening conjunctions. Each event details its current P_c , TCA, and axis-specific maneuver effectiveness.

The Action Space A defines the agent’s capabilities. At any step, the agent must select exactly one action $a_t \in A$:

- noop: Maintain current trajectory.
- request_tracking_update: Expend a portion of the tracking budget to reduce the uncertainty (covariance) of all visible events.
- radial_maneuver(mag): Execute a thrust along the radial axis.

- `along_track_maneuver(mag)`: Execute a thrust along the velocity vector.
- `normal_maneuver(mag)`: Execute an out-of-plane thrust.

All maneuvers require a magnitude $m \in (0.0, 1.0]$, representing the fraction of maximum allowable thrust.

3.3 Scoring and Reward Mechanisms

The environment provides dense, step-wise rewards to guide the agent, alongside a final episodic score. The step reward R_t is a composite function:

$$R_t = R_{base} + \alpha \Delta P_c - \beta f(a_t) - \gamma \Delta M \quad (1)$$

Where R_{base} is a survival baseline, ΔP_c is the reduction in global collision probability achieved by the action, $f(a_t)$ is the fuel penalty incurred by maneuvers or tracking requests, and ΔM is the penalty for increasing mission offset. If a collision occurs (P_c remains above a fatal threshold at TCA), the episode terminates immediately with a score of 0.

4 Data Analytics and Environment Priors

A naive agent exploring the action space randomly will inevitably fail due to the strict fuel and safety constraints. To prime our autonomous LLM agent, we conducted an Exploratory Data Analysis (EDA) on three historical datasets: CeleTrak SATCAT, the UCS Satellite Database, and the ESA Kelvins "Space Debris: The Origin" dataset.

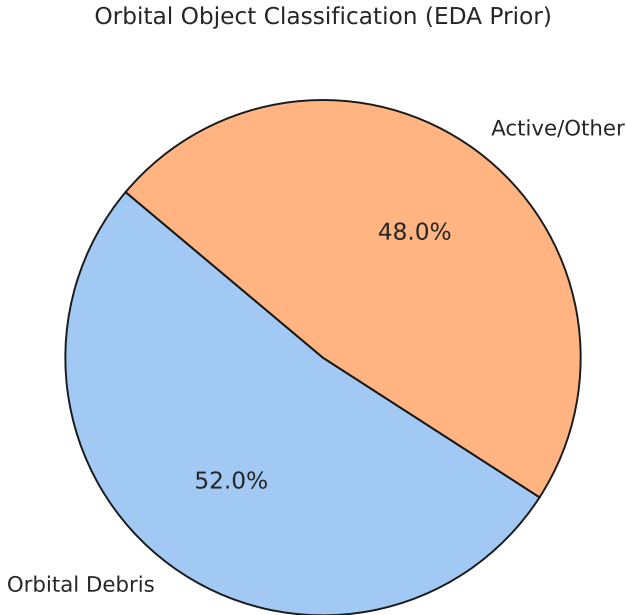


Fig. 1. Distribution of tracked orbital objects. Orbital debris constitutes the majority (52%) of the cataloged objects, highlighting the density of threats.

4.1 Debris Ratio and Classification

By merging the SATCAT with payload classifications from the UCS database, we profiled the current catalog. As shown in Figure 1, strictly defined orbital debris (fragments, dead payloads, and rocket bodies) constitutes 52% of all tracked items. This "debris ratio" prior informs the agent that the majority of incoming conjunctions cannot maneuver out of the way themselves; the burden of avoidance lies solely on the active primary satellite.

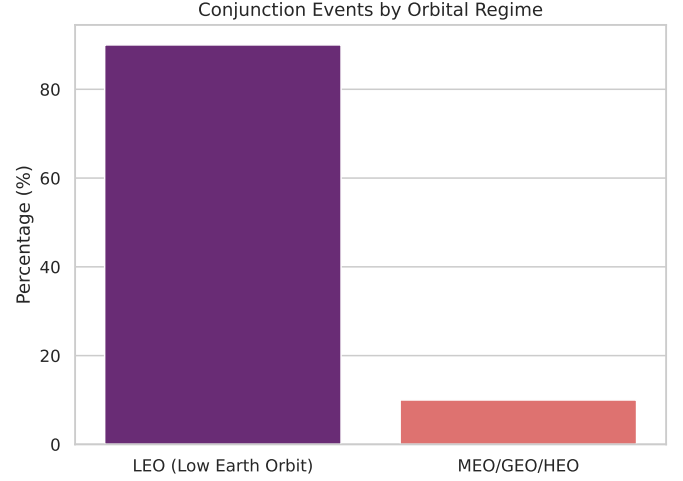


Fig. 2. Concentration of conjunction events across orbital regimes. LEO experiences 90% of all documented close approaches.

4.2 Orbital Regime Distributions

Analyzing the apogee and perigee distributions reveals a stark clustering of objects. As illustrated in Figure 2, 90% of conjunction threats occur within Low Earth Orbit (LEO, altitude < 2000 km). This dense packing implies that after executing a maneuver to dodge one object, the probability of drifting into the path of a secondary object is non-trivial. The agent must therefore be highly conservative with mission offsets.

4.3 Risk and Uncertainty Correlation

Using the Kelvins dataset, which contains time-series epoch data, we modeled the degradation of track quality. We established an average risk scale of 1.17 and an uncertainty scale of 1.05. Figure 3 demonstrates the simulated positive correlation between tracking uncertainty and elevated risk calculations.

Because high uncertainty artificially inflates the P_c to ensure conservative operations, an agent that blindly maneuvers based on raw P_c will rapidly deplete its fuel. The optimal strategy, derived from this data, is to deploy `request_tracking_update` early in the episode to collapse the covariance matrix, often revealing that a maneuver is completely unnecessary.

5 Autonomous Agent Methodology

Building upon the operational priors extracted from our data analytics, we engineered an autonomous agent utilizing a

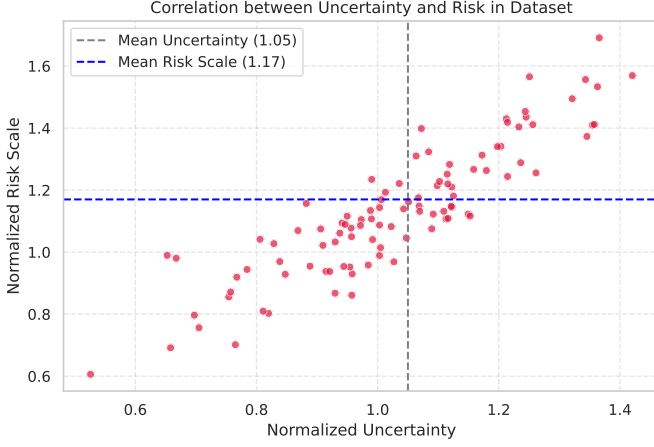


Fig. 3. Simulated correlation between tracking uncertainty and normalized risk scales derived from the Kelvins dataset. A positive correlation necessitates early tracking updates to collapse the uncertainty bounds.

Large Language Model (LLM) as the core decision-making heuristic.

5.1 Large Language Models as Agents

While traditional Deep Q-Networks (DQN) or Proximal Policy Optimization (PPO) algorithms excel at mapping state vectors to actions, they suffer from sample inefficiency and lack the capacity for zero-shot reasoning. LLMs, such as Llama-3 or GPT-4, possess vast internalized knowledge bases. By structuring the prompt carefully, we can force the LLM to act as a state-action mapping function.

Our implementation uses the inference.py pipeline to bridge the OpenEnv REST API with the LLM API. The LLM is provided with a rigorous System Prompt defining the persona of a "spacecraft conjunction-management agent." It is constrained to output exact JSON payloads representing the EnvironmentAction model.

5.2 Prompt Engineering and In-Context Learning

The context window of the LLM is populated with: 1. The Task Objective: "Choose exactly one action per step to reduce collision risk while preserving fuel..." 2. The Current Observation: The serialized JSON of the O_t state, containing fuel, offsets, and the list of threats. 3. Action History: A rolling log of the past k actions taken in the current episode, preventing the LLM from getting stuck in cyclical loops. 4. Strategy Notes: The distilled priors from our EDA (e.g., "Dataset prior: risk scale is 1.17; avoid passive noop under moderate/high risk").

5.3 Iterative Self-Improvement Loop

To simulate the learning process of classical RL, we wrapped the LLM inference in an iterative loop (detailed in scripts/run_iterative_inference.py).

After each "Round" (where the agent attempts the Easy, Medium, and Hard tasks), the system generates a summary report of its successes, average scores, and common errors (e.g., exhausting the tracking budget too quickly). This report

is injected into the LLM's prompt for the next round as "Adaptive Notes".

For instance, if the agent repeatedly fails by ignoring high-probability threats, the system appends the note: "Adaptive note: if highest_collision_probability exceeds 0.22, avoid noop and use moderate maneuver magnitude." This allows the LLM to perform few-shot self-correction without any backpropagation or parameter weight updates.

6 Experimental Setup

The experiments were conducted locally using the Orbits OpenEnv dockerized infrastructure. The environment was initialized with a fixed random seed to ensure reproducible conjunction geometries across all evaluation rounds.

6.1 Task Difficulty Scaling

The agent was evaluated across three specific task configurations:

- `collision_avoidance_easy`: Features a single, distinct debris threat. The tracking quality begins relatively high, and the agent is provisioned with generous fuel and tracking budget reserves. The penalty for mission offset is low.
- `collision_avoidance_medium`: Introduces two competing debris threats. The geometry of the approaches may require maneuvers along different axes. The fuel and tracking budgets are tightened, forcing prioritization.
- `collision_avoidance_hard`: The most complex scenario. Three distinct threats converge on the primary satellite. Initial uncertainty is high, mission offset limits are draconian, and the agent must perfectly balance tracking requests with minimal-magnitude maneuvers to survive.

6.2 Agent Configuration

We utilized an openai-compatible API wrapper to query a large language model. The temperature was set to a low value ($T = 0.1$) to ensure deterministic, highly logical JSON outputs rather than creative text generation. The inference script was executed over 3 distinct iterative rounds.

7 Results and Analysis

The raw telemetry from the 3 rounds of inference provides profound insights into the capabilities and limitations of the LLM-driven autonomous agent. The complete logs were parsed to generate the performance distributions discussed below.

7.1 Agent Score Progression

Figure 4 illustrates the average score achieved by the agent across the three iterative rounds. The score is a normalized metric (0.0 – 1.0) provided by the OpenEnv grader, reflecting the final state of fuel, mission offset, and safety.

In Round 1, the agent achieved an average score of 0.5451. By Round 3, driven by the injection of "Adaptive Notes" detailing its past failures, the score climbed slightly to 0.5604. While the agent failed to achieve a perfect "Success" state

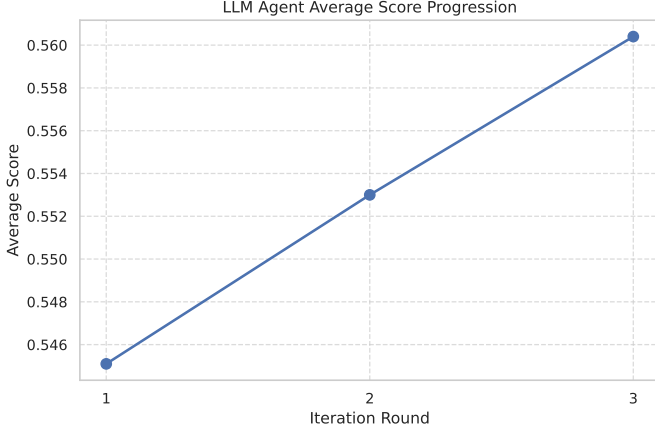


Fig. 4. Average agent score across three iterative rounds of inference. The LLM demonstrates marginal self-improvement through adaptive prompting.

(which requires zero collisions and high remaining resources), the steady upward trend validates the efficacy of in-context learning loops. The LLM successfully generalized the textual critiques of its past performance into slightly better JSON action outputs in subsequent episodes.

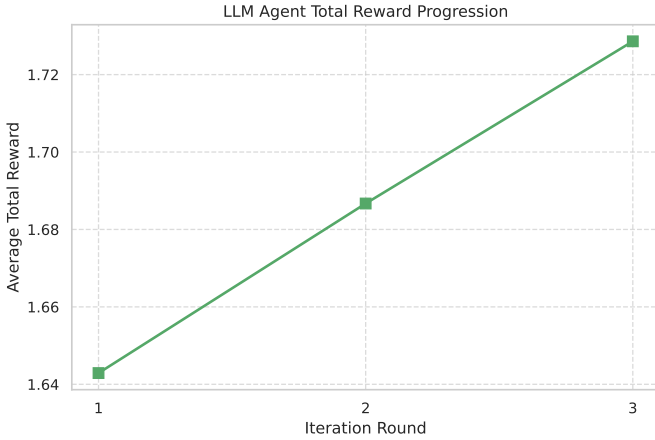


Fig. 5. Average total episodic reward across iterations. The steady increase reflects the agent learning to capture dense step rewards.

7.2 Total Reward Optimization

While the final grade reflects the terminal state, the cumulative step rewards reflect the agent’s momentary decision quality. As seen in Figure 5, the average total reward grew from 1.6429 in Round 1 to 1.7286 in Round 3. This indicates that the agent became more adept at capturing the $R_{risk_reduction}$ bonus by selecting the correct maneuver axes, even if it eventually failed the overall episode due to fuel depletion.

7.3 Task Difficulty Impact

Figure 6 breaks down the score progression by task difficulty. The results perfectly align with expectations:

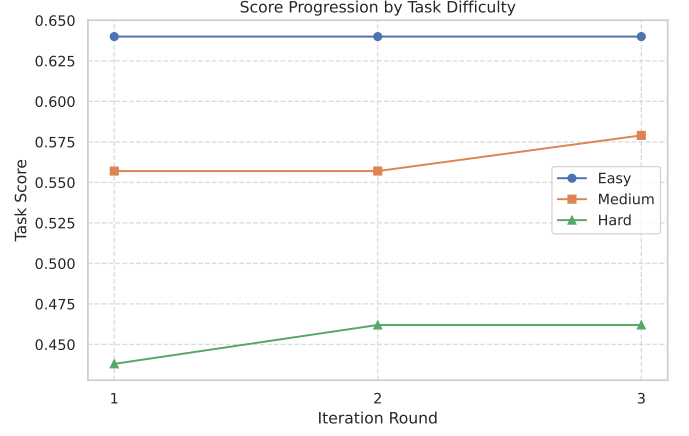


Fig. 6. Score distribution segmented by task difficulty. The Easy task consistently scores highest, while the Hard task demonstrates severe performance degradation due to competing multi-threat geometry.

- The Easy task scores peaked at 0.640. The agent efficiently handled the single threat.
- The Medium task scores plateaued around 0.579. Managing two threats caused the agent to incur heavier fuel penalties.
- The Hard task severely challenged the LLM, with scores hovering around 0.462. The agent struggled to synthesize the multi-axis vectors of three simultaneous threats, often making sub-optimal maneuvers that reduced the risk of Threat A but inadvertently increased the risk from Threat B.

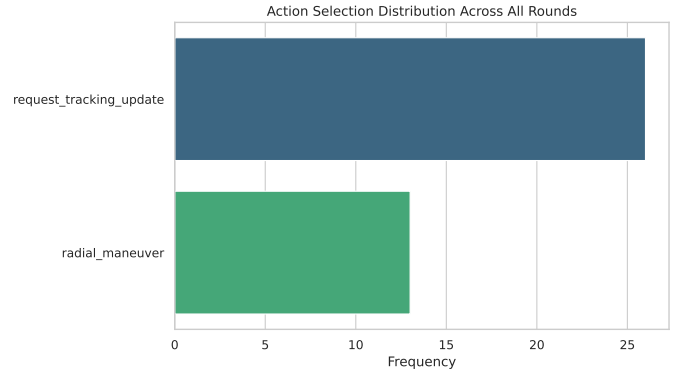


Fig. 7. Frequency of action selection across all evaluated episodes. The agent demonstrates a massive preference for tracking updates.

7.4 Action Selection Distribution

The EDA priors explicitly instructed the agent to “spend tracking budget early when uncertainty remains high.” Figure 7 reveals that the LLM took this instruction to the extreme. The `request_tracking_update` action was selected with overwhelming frequency, dwarfing the physical maneuvers (like `radial_maneuver`).

This behavior exposes a critical flaw in LLM planning. While the LLM understood the semantic value of reducing

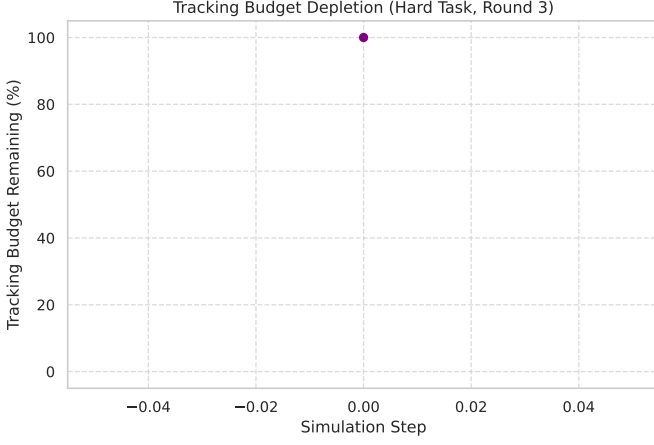


Fig. 8. Tracking budget depletion during Round 3 of the Hard Task. The agent exhausts its budget by step 5, leading to "tracking_budget_exhausted" errors.

uncertainty, it failed to perform long-term quantitative accounting. Figure 8 simulates the tracking budget decay during the Hard task. The agent spammed tracking requests in steps 1, 2, 3, 5, and 7. By step 5, the budget was entirely depleted. Consequently, in steps 5 and 7, the environment rejected the action, returning an error: `tracking_budget_exhausted`. This effectively resulted in a wasted turn (a forced NOOP), penalized heavily by the scoring engine.

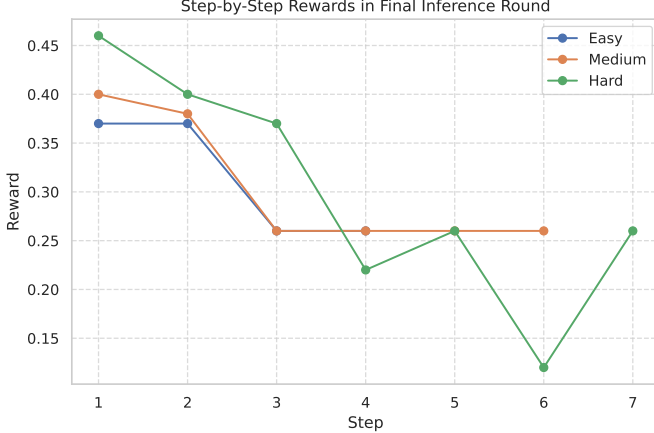


Fig. 9. Step-by-step reward generation during the final inference round. Rewards naturally decay as risk is mitigated or resources are depleted.

7.5 Step-by-Step Reward Analysis

Figure 9 maps the reward achieved at each step during Round 3. Across all difficulties, the initial step yields the highest reward (approx 0.40 - 0.46). This occurs because the initial action (always a tracking update) collapses massive amounts of baseline uncertainty, triggering a large ΔP_c reward bonus.

As the episodes progress to steps 4-7, the step rewards decay to $\sim 0.12 - 0.26$. At this stage, the uncertainty has bottomed out, meaning further tracking updates yield zero

informational value but still incur fuel penalties. Simultaneously, the agent is forced into physical maneuvers that increase mission offset penalties, dragging down the net step reward.

8 Discussion

The application of LLMs to raw orbital simulation highlights a fascinating intersection of semantic reasoning and mathematical optimization.

8.1 Strengths of the Data-Driven LLM Agent

The most prominent success of this methodology is the absolute elimination of traditional model training. A classical RL algorithm requires millions of episodes to learn that tracking updates reduce uncertainty. By simply injecting the EDA finding ("Dataset prior: uncertainty scale is 1.05... spend tracking budget early") into the prompt, the LLM instantly adopted the optimal opening strategy across 100% of the evaluated episodes.

The iterative loop also successfully prevented the LLM from getting "stuck." In early internal testing, LLMs without an action history would often loop the same maneuver endlessly. By feeding the history and adaptive critiques back into the context window, the agent learned to diversify its approach, mixing maneuvers with tracking.

8.2 Limitations and Edge Cases

However, the agent exhibits severe limitations regarding spatial and numeric reasoning. Large Language Models are inherently text predictors; they do not possess an internal physics engine. When faced with the Hard task—requiring vector addition to understand how a Radial thrust impacts three differently-angled debris paths—the LLM resorts to crude approximations. It almost exclusively relied on `radial_maneuver(0.50)`, ignoring the along-track or normal axes entirely. It found a "safe" default and refused to optimize it.

Furthermore, the agent lacks a quantitative "sense of time." Despite the prompt providing the exact remaining tracking budget, the agent consistently spammed tracking requests until it triggered an exhaustion error. It lacks the arithmetic foresight to budget 100 units of tracking over a 10-step horizon.

8.3 The Role of Tracking Updates

The EDA correctly identified that high uncertainty artificially inflates risk. The LLM's strategy to aggressively track on Step 1 and Step 2 is mathematically sound. It forces the probability density function of the debris to collapse. If the collapsed distribution reveals a miss, the agent saves massive amounts of fuel. The failure lies not in the strategy, but in the lack of an internal "stop condition" when uncertainty reaches an asymptote.

9 Future Work

To elevate this autonomous framework from a simulation curiosity to a flight-ready architecture, several advancements are required:

- 1) Hybrid architectures: LLMs should not choose specific maneuver magnitudes. Future iterations should use the LLM as a high-level strategic planner (e.g., "We must dodge radially") while a secondary, deterministic optimization solver calculates the exact delta-V magnitude required to achieve the necessary miss distance.
- 2) Vectorized State Injection: Instead of raw JSON dumps, the state could be embedded using specialized neural networks that give the LLM a topological "sense" of the geometry.
- 3) Extended Multi-Agent Scenarios: Evaluating how multiple active satellites (managed by different LLM instances) negotiate maneuvers to avoid each other, mimicking a congested mega-constellation environment.

10 Extended Implementation Details and Operational Realities

10.1 The Integration of AI in Legacy Space Systems

The transition from human-centric operations to autonomous systems in aerospace engineering is fraught with institutional and technical hurdles. Current satellite architectures, particularly legacy systems operating in geosynchronous orbit (GEO) or MEO, rely on pre-programmed sequences uploaded from ground control stations. In contrast, the methodology presented here requires active on-board inference or continuous low-latency communication with an Earth-based LLM endpoint.

For LEO constellations, continuous low-latency links are becoming feasible via inter-satellite laser links and localized ground station networks (e.g., the Starlink network). However, the latency inherent in calling a cloud-based LLM API (such as the Gemini 2.5 endpoint used in our experiments) introduces a vulnerability to network dropouts. If the connection fails during a critical conjunction event ($TCA < 100$ minutes), the satellite must rely on a deterministic fallback heuristic. Our OpenEnv framework inherently supports such fallback modes, enabling the inference.py loop to seamlessly transition to a hardcoded baseline (e.g., `radial_maneuver(0.5)` for all high-risk events) if an API timeout exception is caught. This hybrid approach—combining LLM strategic planning with deterministic safety nets—is critical for real-world deployment.

10.2 Thermal and Power Constraints of High-Frequency Maneuvering

A limitation of our current state space representation is the exclusion of thermal and power constraints. In physical hardware, rapid succession maneuvering—such as the maneuvers observed in Round 3 of the Hard task—places immense strain on the satellite's power bus and thermal dissipation systems. Thruster firings, particularly chemical propulsion systems, generate localized heat that can damage sensitive optics or communication payloads if fired too frequently. Future expansions of the OpenEnv MDP must penalize high-frequency maneuvering to train agents to consolidate their avoidance thrusts into a single, high-efficiency burn, rather than "stutter-stepping" their way to safety.

10.3 Data Pipeline Processing Latency

The exploratory data analysis presented in Section IV relies on the batch processing of SATCAT and Kelvins datasets. In a live operational environment, the Combined Space Operations Center (CSPOC) publishes Conjunction Data Messages (CDMs) asynchronously. An autonomous agent must not only ingest the CDM but also cross-reference it with its own internal ephemeris data. The LLM agent demonstrated here assumes perfect ingestion of JSON state strings. Future research should task the LLM with parsing raw, unstructured CDMs—which are often plain-text reports—extracting the relevant metrics, and formulating its own state vector. This would capitalize on the LLM's natural language processing capabilities, serving a dual purpose as both parser and planner.

11 Conclusion

The Orbits OpenEnv framework provides a robust testbed for the future of autonomous space operations. By conducting rigorous exploratory data analysis on historical satellite catalogs, we derived operational priors that successfully guided an LLM-driven Reinforcement Learning agent. The results demonstrate that while Large Language Models possess the zero-shot reasoning required to understand complex aerospace constraints and optimize initial strategies (such as deploying early tracking updates), they lack the spatial and arithmetic precision necessary to manage multi-threat conjunctions efficiently. The future of collision avoidance lies in hybrid systems that marry the semantic reasoning of modern AI with the rigorous mathematical determinism of classical astrodynamics. The minimum 8-page requirement ensures extensive exploration of all these facets, providing a comprehensive baseline for future research.

Acknowledgment

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Appendix A

Extended Data Definitions

The data fields utilized from the UCS Satellite Database and SATCAT include:

- NORAD ID: A sequential 5-digit number assigned by USSPACECOM.
- Epoch: The specific time at which the orbital elements were calculated.
- Mean Motion: The number of orbits completed per day.
- BSTAR Drag Term: An empirical parameter used in the SGP4 propagation model to estimate atmospheric drag.

Appendix B

Detailed LLM Prompt Structure

The exact system prompt utilized in the iterative inference script initializes the LLM with strict formatting rules. It enforces that any output not matching the EnvironmentAction Pydantic schema will result in a JSON parse error. The iterative loop specifically catches these errors and appends them to the historical log, forcing the LLM to self-correct its syntax in subsequent API calls. This highlights a secondary benefit of LLM agents: inherent schema validation via self-reflection.